

Response to Reviewers

Manuscript PONE-D-26-06954R1

Evaluating the effectiveness of open-weight LLMs for multi-domain Bengali text classification with zero-shot and few-shot prompting

Dear Dr. Bhowmick,

Thank you for the opportunity to revise our manuscript, and please convey our thanks to both reviewers for their careful reading. Reviewer #2 raised three remaining points and Reviewer #3 asked for a focused revision of the scientific positioning and the statistical analysis. We have addressed every point, as detailed below. In the file labelled “Revised Manuscript with Track Changes” all new or modified text is shown in blue; page and line numbers below refer to that file. The file labelled “Manuscript” contains the same text without markup.

Two changes affect the submission metadata and we would be grateful if they could be reflected in the system: (1) following Reviewer #3’s suggestion, the term “open-source” has been replaced by “open-weight” throughout the manuscript, including in the title; (2) five references were added (statistical methods), so the reference list now contains 60 entries.

We did not receive any recommendation to cite specific previously published works, and no attachment files accompanied the reviews.

Reviewer questionnaire

1. Is the manuscript technically sound, and do the data support the conclusions? Reviewer #2: Yes. Reviewer #3: Partly.
2. Has the statistical analysis been performed appropriately and rigorously? Reviewer #2: Yes. Reviewer #3: No.
3. Have the authors made all data underlying the findings fully available? Reviewer #2: Yes. Reviewer #3: Yes.
4. Is the manuscript presented in an intelligible fashion and written in standard English? Reviewer #2: Yes. Reviewer #3: Yes.

The “Partly” and “No” of Reviewer #3 concern the statistical analysis and the strength of some conclusions. These are addressed under Reviewer #3 below; in short, all six models are now compared with a paired multi-model analysis with multiple-comparison correction, the few-shot effect is tested rather than assumed, and the causal language about model size has been removed.

Reviewer #2

The authors have responded appropriately to my comments and explain both how they addressed them and how they highlighted them in the relevant section of the article. I have only three important points left to address.

We thank the reviewer for the positive assessment and for the three remaining points.

Point 1

Reviewer comment: *In the current Tables 8 and 9, all values related to “Hate Speech Detection” and “Fake News Detection” (for all models) have not changed from the original version. It appears that these values still correspond to the F1 scores, but not to the newly calculated Macro-F1 scores.*

Response: The values in these two blocks are indeed Macro-F1 scores. They did not change between the two versions for a mathematical reason: the evaluation sets of the two binary tasks are exactly balanced (500 hateful and 500 non-hateful texts; 254 fake and 254 real articles). When the classes have equal

support, the support-weighted F1 that we reported originally and the Macro-F1 are the same number, so the values for these two tasks are identical under both definitions. We now say this explicitly in the Evaluation metrics section. To make sure that the tables are fully consistent, we re-derived every value in both tables from the released prediction files. This confirmed all Hate Speech and Fake News values and also revealed a transcription error in the Emotion Recognition Macro-F1 column of the previous revision: the zero-shot column contained the few-shot values, and the few-shot column had duplicated the Hate Speech values. The corrected values are LLaMA-3.2-3B 45.88 / 46.35, Mistral-V3-7B 36.08 / 45.18, DeepSeek-R1-8B 44.08 / 45.85, Phi-4-14B 59.89 / 57.90, Gemma-2-27B 61.80 / 63.01 and Qwen-2.5-72B 61.91 / 66.80 (zero-shot / few-shot). They agree with the confidence-interval table that already appeared in the previous version. All sentences that quoted the wrong numbers have been corrected, and eight further cells of the zero-shot table that differed slightly from the prediction files were corrected at the same time. Please note that the tables are now numbered 10 and 11 because two tables were added earlier in the Results section.

Changes made in the manuscript:

- Note on balanced evaluation sets and the equality of Macro-F1 and weighted F1: Evaluation metrics, p. 25, line 604.
- Corrected Emotion Recognition Macro-F1 column (and eight further cells) in Table 10 (p. 33) and Table 11 (p. 35); corrected cells are shown in blue.
- Corrected numbers in the text: Abstract (p. 1); Zero-shot performance, p. 32, line 765; Few-shot performance, p. 34, line 790 and p. 34, line 813; Discussion, p. 47, line 1115; Conclusion, p. 54, line 1322.

Point 2

Reviewer comment: *The figures showing the class-wise performance analysis (Figs. 11, 12, 13, 14) present F1-score percentages, rather than Macro-F1 scores, as shown in Tables 8 and 9.*

Response: The class-wise figures show the F1-score of each class separately, which is what a class-wise analysis is meant to do; the Macro-F1 in the tables is the unweighted mean of exactly these per-class values. For example, the zero-shot sentiment F1-scores of Gemma-2-27B are 71.1 (negative), 39.5 (neutral) and 65.4 (positive), whose mean is 58.7, the Macro-F1 of 58.66% given in Table 10. We have added a sentence at the beginning of the class-wise section and a note in each of the four figure captions to make this relation explicit. The figures themselves were regenerated only to harmonise the model names on the axes with the rest of the paper (see Reviewer #3, point 8); the plotted values are unchanged. Because a duplicated copy of Fig 1 was removed from the source file, these figures are now numbered 10 to 13.

Changes made in the manuscript:

- Explanatory sentence at the start of the class-wise section: p. 35, line 821.
- Note added to the captions of Fig 10 (p. 36), Fig 11 (p. 37), Fig 12 (p. 38) and Fig 13 (p. 39).

Point 3

Reviewer comment: *The sections titled “LLMs performance across sentiment categories” and “LLMs performance across emotion categories” contain a lot of repeated text (copied and pasted). They need to be reviewed and corrected.*

Response: We apologise for this oversight, which was introduced when the tracked changes of the previous revision were merged. In the sentiment subsection an entire paragraph was duplicated and one sentence appeared four times; in the emotion subsection one sentence appeared three times. All duplicates have been removed, and both subsections were re-read in full. While doing so we also corrected four statements that did not agree with the plotted values (the size of the improvements for Sadness and Surprise, the direction of the few-shot change for Phi-4-14B on hate speech and for DeepSeek-R1-8B on fake news, and one per-class value for Gemma-2-27B).

Changes made in the manuscript:

- Duplicated paragraph and sentences removed in “LLMs performance across sentiment categories” (p. 35, line 825 ff.) and “LLMs performance across emotion categories” (p. 36, line 844 ff.).
- Corrected statements: p. 36, line 853; pp. 37–38, lines 878–886; p. 38, line 899.

Reviewer #3

The paper evaluates six open-source LLMs across four Bengali classification tasks using zero- and five-shot prompting. My recommendation: Major Revision – but the paper is potentially publishable after a focused methodological/statistical revision. [...] In summary, next revision primarily on scientific positioning and statistical rigor rather than adding many more experiments.

We thank the reviewer for this constructive review, which we found very helpful. As suggested, the revision concentrates on scientific positioning and statistical rigour. No new model runs were required: all additional analyses were computed from the prediction files that were already released with the paper, and the analysis scripts have been added to the public repository. The fourteen numbered points are answered in order, followed by the three general remarks (multi-model analysis, “understanding”, and benchmark versus real-world effectiveness).

Point 1 — Remove/soften causal claims about model size

Reviewer comment: *The paper cannot really establish that “model size” causes better performance. [...] Remove/soften causal claims about model size.*

Response: We agree. Every statement that attributed performance to model size as a cause has been rewritten as an association, and the reason is stated: each parameter scale is represented by a single model family, so size is confounded with pre-training corpus, tokenizer, instruction tuning and, for Qwen-2.5-72B, quantization. We also point out that the ranking is not monotonic in size (Gemma-2-27B outperforms the 72B Qwen model on two tasks; LLaMA-3.2-3B is indistinguishable from Phi-4-14B on zero-shot fake news detection; the 7B Mistral and 8B DeepSeek models swap ranks between tasks).

Changes made in the manuscript:

- Abstract (p. 1): conclusion sentence rewritten.
- Contribution 3: p. 6, lines 103–106.
- Zero-shot performance: p. 33, lines 775–785.
- Few-shot performance: p. 35, lines 817–818; Error analysis: p. 40, line 946.
- Discussion: pp. 45–46, lines 1078–1097 and p. 46, line 1097 ff.; Implications: p. 50, line 1212.
- Conclusion: p. 53, line 1312.

Point 2 — Remove “always/constantly” language

Reviewer comment: *The claim “larger models constantly surpass smaller ones across all tasks” is too strong. [...] Remove “always/constantly” language.*

Response: The quoted sentence (Conclusion) and its counterparts in the Abstract (“always performed better”) and Discussion (“uniformly superior on all tasks”, “consistently”) have been removed or replaced by statements that are supported by the tests: the two largest models obtain the highest scores on every task, but the ordering below them is task-dependent and several neighbouring models cannot be separated statistically.

Changes made in the manuscript:

- Abstract (p. 1).
- Discussion: p. 45, line 1078 ff.
- Conclusion: p. 53, line 1312.

Point 3 — Clarify that only Qwen vs Gemma were statistically compared

Reviewer comment: *The study evaluates six models, but inferential testing focuses only on the two strongest models. [...] Clarify that only Qwen vs Gemma were statistically compared.*

Response: We now state explicitly, both in the Methods and where the results are reported, that the permutation tests concern only the Qwen-2.5-72B versus Gemma-2-27B pair, and we list this in the Limitations. More importantly, we have added the multi-model analysis that the reviewer requested (see the general remark below), so that statements about the remaining models are now supported by tests.

Changes made in the manuscript:

- Statistical significance testing (Methods): p. 27, line 650.
- Statistical significance analysis (Results): p. 29, line 698.
- Limitations: p. 52, line 1290.

Point 4 — Address multiple-comparison correction

Reviewer comment: *Multiple-testing correction is missing. [...] Address multiple-comparison correction.*

Response: All families of tests are now corrected with the Holm–Bonferroni step-down procedure. The 16 p-values of the original permutation tests were adjusted as one family and are reported next to the raw values in Table 6; after correction, five of the eight originally significant tests remain significant (hate speech detection under few-shot prompting for both metrics, hate speech detection under zero-shot prompting for Macro-F1, emotion recognition under few-shot prompting for accuracy, and sentiment analysis under zero-shot prompting for accuracy), and the text has been rewritten accordingly. The new pairwise McNemar tests (15 pairs per dataset and setting) and the 24 zero-shot versus few-shot tests are Holm-corrected as well. We also clarify that the permutation test is two-sided and how p-values estimated as zero are treated. References for Cochran’s Q, the McNemar test and the Holm procedure, together with the standard references on statistical comparison of classifiers (Dietterich 1998; Demšar 2006), have been added.

Changes made in the manuscript:

- Methods: p. 27, line 639; pp. 27–28, lines 651–661.
- Table 6 (p. 29): raw and Holm-adjusted p-values; interpretation rewritten at p. 29, line 686 ff.
- Table 7 (p. 31) and Table 8 (p. 31): Holm-corrected pairwise and zero- vs few-shot tests.
- References 51–55 added (Cochran 1950; McNemar 1947; Dietterich 1998; Holm 1979; Demšar 2006).

Point 5 — Qualify the five-shot result because only one demonstration set/seed was used

Reviewer comment: *One fixed set of five demonstrations is not enough to robustly measure the effect of few-shot prompting. [...] Qualify the five-shot result because only one demonstration set/seed was used.*

Response: We agree, and we now qualify the few-shot results throughout. The Prompting strategies section describes the single demonstration set that was used (including its label composition, for example three hate and two not-hate examples, and no anger example for emotion recognition) and states that the results quantify the effect of this particular set rather than of five-shot prompting in general. We also tested, for each model and task, whether the few-shot predictions differ from the zero-shot predictions on the same items (exact McNemar test, Holm-corrected over the 24 comparisons; new Table 8). Only four of the 24 changes are significant, three improvements and one degradation, all for the two smallest models; none of the changes for Phi-4-14B, Gemma-2-27B or Qwen-2.5-72B is significant. The Results, Discussion, Limitations, Abstract and Conclusion have been rewritten so that the few-shot findings are described as effects of one fixed demonstration set on classification performance, and RQ3 no longer presupposes an improvement. In line with the reviewer’s remark that the revision should focus on positioning rather than on additional experiments, we did not run further demonstration sets; this is stated as a limitation and as future work.

Changes made in the manuscript:

- Prompting strategies: pp. 22–23, lines 525–533.
- Table 8 (p. 31) and accompanying text: pp. 30–31, lines 729–742.
- Few-shot performance: p. 34, line 793, p. 34, line 797, p. 34, line 805.
- Discussion: p. 48, line 1160 ff.; Implications: p. 50, line 1219; Limitations: p. 52, lines 1263–1269.
- RQ3: p. 5, line 82; Contribution 4: p. 6, lines 108–112; Abstract (p. 1); Conclusion: p. 54, line 1323.

Point 6 — Clarify that fake-news downsampling changes the evaluation distribution

Reviewer comment: *Clarify that fake-news downsampling changes the evaluation distribution.*

Response: The Datasets section now states that fabricated articles constitute only about 1.3% of the source corpus but 50% of our evaluation set, that the reported scores therefore describe performance under a balanced class prior, and that the precision of the fake class in particular would be much lower at the natural prevalence. The same point is made in the Discussion and listed in the Limitations.

Changes made in the manuscript:

- Datasets and tasks (Fake news detection): pp. 12–13, lines 279–283.
- Discussion: p. 48, line 1154; Limitations: p. 52, line 1279.

Point 7 — Replace “understanding” claims with “classification performance” claims

Reviewer comment: *There is a conceptual problem with the claim that few-shot prompting improves Bengali understanding – I recommend change to “improve classification performance under the evaluated Bengali task settings.” [...] Replace “understanding” claims with “classification performance” claims where appropriate.*

Response: We adopted the reviewer’s formulation. The contribution that claimed improved “understanding of low-resource languages” now describes changes in classification performance under the evaluated Bengali task settings, and the same replacement was made wherever the results were interpreted as language understanding, contextual understanding, or the ability to “detect” or “identify” phenomena beyond the benchmark.

Changes made in the manuscript:

- Contribution 4: p. 6, lines 108–112; RQ3: p. 5, line 82.
- Zero-shot performance (p. 32); Class-wise analysis: p. 36, line 839; Error analysis: p. 40, line 946; ROC-AUC analysis: p. 45, line 1072.
- Discussion: p. 48, line 1160 ff.

Point 8 — Ensure exact model names/revisions are consistent throughout

Reviewer comment: *The model naming needs absolute consistency. [...] Ensure exact model names/revisions are consistent throughout.*

Response: A paragraph at the beginning of the Methodology now fixes the six short names used in all text, tables and figures and maps each of them to the exact Hugging Face checkpoint listed in the Models subsection. While checking the identifiers against the run scripts we found that the Gemma checkpoint had been listed as the base model (google/gemma-2-27b) although the instruction-tuned model google/gemma-2-27b-it was evaluated; this has been corrected. The remaining inconsistencies (LLaMa/Llama, lower-case suffixes in figure axes, raw internal keys in the ROC legends) were removed: all class-wise, ROC and overall figures were regenerated from the same prediction files with identical model names, and the plotting code, README and job scripts in the repository were aligned with the paper.

Changes made in the manuscript:

- Methodology: pp. 10–11, lines 231–237.

- Models: corrected Gemma identifier and revision statement, p. 13, line 308.
- Figures regenerated with consistent names: Fig 10 (p. 36)–Fig 13 (p. 39), Fig 16 (p. 46), Fig 17 (p. 46); plotting script and README in the repository.

Point 9 — Strengthen the limitations around model-family confounding

Reviewer comment: *One particularly important issue: “model size” and “multilingual pretraining” are being conflated. [...] Strengthen the limitations around model-family confounding.*

Response: A dedicated paragraph in the Discussion and a new item in the Limitations explain that model size and model family go together in our design (one family per scale, differing in pre-training data and language coverage, tokenizer vocabulary, instruction tuning and quantization), that no within-family size series was evaluated, and that statements about size are therefore associational. We also note the two observations in our own data that argue against size as the sole explanation (Gemma-2-27B versus Qwen-2.5-72B; DeepSeek-R1-8B versus LLaMA-3.2-3B).

Changes made in the manuscript:

- Discussion: p. 46, line 1097 ff.
- Limitations: p. 52, line 1284 ff.
- Contribution 3: p. 6, line 103; Zero-shot performance: p. 33, line 775.

Point 10 — Add quantitative error-category analysis

Reviewer comment: *Error analysis is much better, but still needs quantitative depth. [...] Add quantitative error-category analysis.*

Response: The error-analysis section was renamed “Qualitative and quantitative error-category analysis”. It now states the basis of the qualitative observations (a random sample of 15 misclassified items per task) and follows each of the four observations with a quantitative check computed over all predictions of all six models: confusion-pair shares (e.g., disgust predicted as anger accounts for 26.2% of all emotion errors; positive items predicted as neutral for 29.6% of all sentiment errors), the false-negative/false-positive asymmetry in hate speech detection (66.9–84.1% false negatives for the four smaller models under zero-shot prompting, 61.2% false positives pooled under few-shot prompting), the error rate on items with an explicit negation marker (6–13 points higher in emotion recognition, but no difference in sentiment analysis), and the surface cues of missed fabricated articles (numerals are more frequent; length and quotation marks are not). The counts derived from the confusion matrices were also added to the “Best performing models” and “Worst performing models” subsections, and the Discussion uses them. Where a category could not be quantified without annotation (sarcasm), we say so. The analysis script and its output files were added to the repository.

Changes made in the manuscript:

- Qualitative and quantitative error-category analysis: pp. 41–44, lines 976–1041.
- Best performing models: p. 39, line 923, p. 39, line 932, p. 40, line 939; Worst performing models: p. 41, line 957, p. 41, line 964.
- Discussion: p. 47, line 1116, p. 47, line 1132, p. 48, line 1143, p. 48, line 1154.

Point 11 — Discuss English versus Bengali prompting as a limitation

Reviewer comment: *Discuss English versus Bengali prompting as a limitation.*

Response: The Prompting strategies section now notes that the language of the instructions was not varied and that the results describe performance under English-instructed prompting only; the Limitations discuss the possible interaction of Bengali or bilingual instructions with each model’s instruction-tuning data and recommend treating the prompt language as an experimental factor.

Changes made in the manuscript:

- Prompting strategies: p. 22, line 501.
- Limitations: p. 52, line 1260.

Point 12 — Consider “open-weight” rather than “open-source”

Reviewer comment: *Consider “open-weight” rather than “open-source.”*

Response: We adopted the suggestion. “Open-source” was replaced by “open-weight” throughout the manuscript, including the title, and a sentence in the Methodology explains the choice: although the weights of all six models are publicly downloadable, several of them (e.g., the Llama and Gemma families) are released under licences that do not meet the Open Source Definition. Apart from the sentence that explains the terminology, the only remaining occurrence of “open-source” is in Table 1, where it describes how the authors of TigerLLM characterised their own model.

Changes made in the manuscript:

- Title (p. 1) and all occurrences in the text.
- Explanation: p. 11, line 234.

Point 13 — Report exact software/model/tokenizer versions

Reviewer comment: *Report exact software/model/tokenizer versions.*

Response: The Experimental setup now reports the software versions used (Python 3.12, PyTorch 2.5.1 with CUDA 11.8, Transformers 4.46.3, torchao 0.6.1, AutoAWQ 0.2.9; scikit-learn and SciPy for metrics and tests), and Table 5 has a corresponding row. The Models subsection states that all six checkpoints are the instruction-tuned versions, that models and tokenizers were loaded from the default revision of the listed repositories, and that each model’s released chat template was used without modification. We also state that each model–task–setting configuration was run once.

Changes made in the manuscript:

- Experimental setup: p. 24, lines 580–582; Table 5 (p. 24), “Software” row.
- Models: p. 13, line 308.

Point 14 — Make the Discussion substantially more analytical and less descriptive

Reviewer comment: *Make the Discussion substantially more analytical and less descriptive.*

Response: The Discussion was rewritten. It now starts from the three findings that survive multiple-comparison correction, examines why the largest models do best in terms of the confounded design, explains the task-level results with the quantified error patterns (neutral drift in sentiment analysis, the disgust–anger boundary and negation in emotion recognition, the prompt-induced label prior in hate speech detection, numeral-rich fabrications in fake news detection), interprets the few-shot results in the light of Table 8, and closes with the distinction between benchmark performance and real-world effectiveness. The descriptive paragraph on “overarching conclusions” was removed. The paragraph on the absence of BanglaBERT and proprietary baselines was kept unchanged.

Changes made in the manuscript:

- Discussion: pp. 45–50, lines 1078–1207.
- Implications for low-resource language NLP: p. 50, line 1212, p. 50, line 1219, p. 51, line 1234.

General remark — “Use a paired multi-model analysis across the six models, followed by corrected pairwise comparisons”

Response: This analysis was added. For each dataset and prompting setting, Cochran’s Q test on the per-item correctness of the six models serves as the omnibus test (all $p < 10^{-25}$), followed by exact McNemar tests for all 15 model pairs with Holm correction. The results are given in the new Table 7 and

the new subsection “Multi-model paired analysis”. The subsection also reports the pairs that cannot be separated after correction, which is what supports the statements about non-monotonic ranking and model-family confounding elsewhere in the paper. Bootstrap confidence intervals of the Macro-F1 are now reported for all six models (Table 9) instead of only for Qwen-2.5-72B and Gemma-2-27B.

Changes made in the manuscript:

- Methods: pp. 27–28, lines 651–661.
- Results: new subsection “Multi-model paired analysis”, pp. 30–31, lines 707–742, with Table 7 (p. 31) and Table 8 (p. 31).
- Table 9 (p. 32): confidence intervals for all six models; text at p. 32, line 750.

General remark — “The paper should distinguish benchmark performance from real-world effectiveness”

Response: This distinction is now made explicitly in the Abstract, in the closing paragraph of the Discussion (small, partly artificially balanced test sets; overlapping confidence intervals; differences in class priors, genres, dialects and adversarial behaviour at deployment), in the Limitations, and in the Conclusion. Statements about what the models can “detect” or “identify” in general were rephrased as benchmark classification performance under the evaluated settings.

Changes made in the manuscript:

- Abstract (p. 1).
- Discussion: pp. 49–50, lines 1191–1207; Limitations: p. 52, line 1279; Conclusion: p. 54, line 1323.

General remark — terminology

Reviewer comment: *The manuscript contains some terminology that should be cleaned up.*

Response: Non-standard terms were replaced (for example “rapid engineering” by “prompt engineering”, “exclusive-use licensed models” by “proprietary models”, “streamlined transformer models” by “fine-tuned encoder-based transformer models”, “virtually grounded prompting” by “few-shot prompting”, and colloquial phrases such as “the bigger guys” by neutral wording). Model names were unified as described under point 8.

Other changes made by the authors

For transparency we list the further changes that are not tied to a specific reviewer comment:

- Two panels of the confusion-matrix figures (Fig 14, panel d, and Fig 15, panel d) had inadvertently reused the image of a neighbouring panel in the previous version; the correct matrices (Qwen-2.5-72B zero-shot emotion recognition; DeepSeek-R1-8B few-shot emotion recognition) were generated from the prediction files and inserted.
- A duplicated copy of Fig 1 in the source file was removed; as a consequence all figure numbers after Fig 1 decreased by one. Two tables (7 and 8) were added, so the zero-shot and few-shot result tables are now Tables 10 and 11.
- One bibliography entry (reference 38, Das et al.) was corrected from a journal to a conference-proceedings entry so that the reference list compiles correctly.
- Two grammatically incomplete sentences in the Zero-shot performance section were corrected.
- The analysis scripts for the multi-model tests, the confidence intervals of all models and the quantitative error analysis, together with their output files, were added to the public repository named in the Data Availability statement, and the README was updated accordingly.

We hope that the revised manuscript now meets the standards of PLOS ONE and thank the editor and the reviewers again for their time and their constructive comments.

Yours sincerely,

Khair Razlan Bin Othman (corresponding author), on behalf of all authors